

The art of information



Class 4: 01/08 (Compression 2)

how much stuff can we
remember?

how much stuff can we
remember?

a countless amount?

just a few?

a countless amount?

long-term memory

just a few?

short-term memory

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THE PSYCHOLOGICAL REVIEW

THE MAGICAL NUMBER SEVEN, PLUS OR MINUS TWO:
SOME LIMITS ON OUR CAPACITY FOR
PROCESSING INFORMATION ¹

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Harvard University

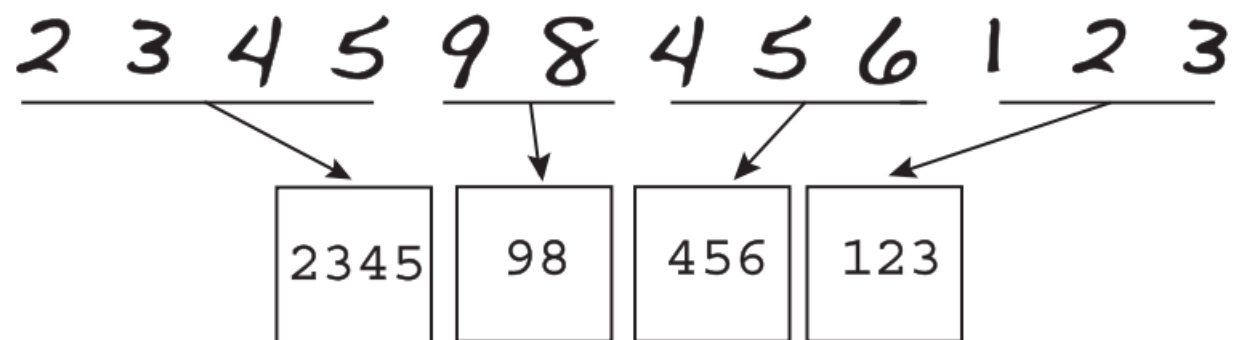
The magical number 4 in short-term memory: A reconsideration of mental storage capacity

Nelson Cowan

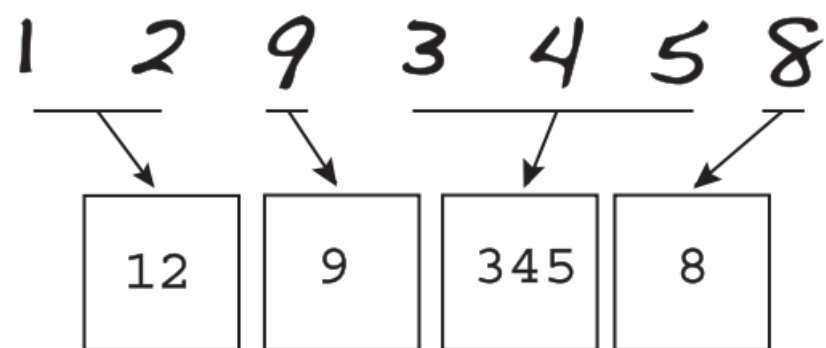
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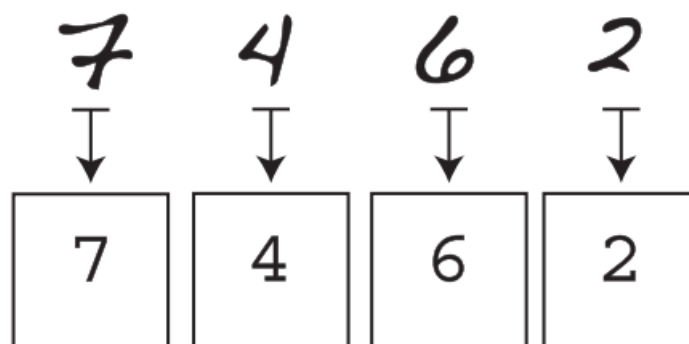
what's the link?
what is a *chunk*?



*Highly patterned sequence—
12 items in 4 chunks*



*Moderately compressible sequence—
7 items in 4 chunks*



*Incompressible sequence—
4 items in 4 chunks*

maximally compressed
code

“An immediate consequence, confirmed in our experiments, is that more compressible material is easier to fit into available encoding space, and less compressible (more complex) material is harder”



ENCODING

Transforming Information into a form that can be stored in memory

STORING

Maintaining the encoded information in memory

RETRIEVING

Re-accessing information from the past which has been encoded and stored.

“This conception compares very closely with common computer data compression techniques such as Lempel–Ziv (e.g. see Ziv & Lempel, 1977), which underlies the widespread compression utility gzip.”



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Encoding



Storage



Retrieval



7 items $\approx$ 4 chunks

7 items $\approx$ 4 chunks

not just in memory, but often in 'reality'



$$p(\text{Continuing run}) = 1/10$$

(really = 0.085)

R possible runs (+1, -1, ...)
everything is a new chunk, unless it's a
run

$N - 1$ tosses of coin p

E(compression ratio):

N

E(complexity)

1.78 (nearly $7/4$!)

linking two impactful
psychological theories
with information theory

Table 1. Pattern-finding by simplicity: a sample of research^a

Cognitive process	Data	Codes	Computer science/ mathematical approaches	Cognitive science applications
Low-level perception	Sensory input/artificially captured images	Filters in early vision	Image compression [25]	Early vision as compression [23,22]
High-level perception	Sensory input/output of early perceptual Processing	Representations of higher level structure	Pattern theory [26]	Principle of economy [1]
Language acquisition	Linguistic input	Representations of language structure	Text compression [28]	Perceptual organization [27,14] Phonological [29] and morphological analysis [30], segmentation [31,24] and grammar induction [32,33]
High-level cognition	High-level representations of knowledge	Similarity, causal relations	Information distance [34]	Similarity as representational distortion [18]
Scientific inference	Scientific data	Theoretical knowledge	Gencompress [35] Machine induction systems [9] Foundations of statistics [10]	Categorization by compression [36] Ockham, Newton Mach's principle of economy [1] Formal measures of simplicity [37,38]

memory can be bound
to context, meaning, ...



vs.



memory can be bound
to context, meaning, ...



memory can be bound
to context, meaning, ...



seems obvious, but it's surprising!

< General

iPhone Storage



iPhone

254.6 GB of 256 GB Used



• Photos • Messages • Apps • iOS • System Data

RECOMMENDATIONS



Optimize Photos

[Enable](#)

Save 229.22 GB - Full resolution photos and videos are safely stored in iCloud and your iPhone will use smaller size versions.

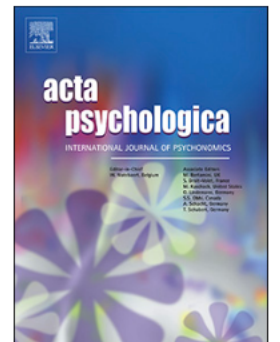
complexity



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The sound of beauty: How complexity determines aesthetic preference

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tomorrow

complexity

The Pudding

WONKY

An exploration of rhythm and grooves that break the rules



By [Michelle Pera-McGhee](#)

START AUDIO STORY



tomorrow

The art of information



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